HMMamba: A Heterogeneous Multibranch Mamba Architecture for Hyperspectral Image Classification

Yongchao Sun, Jing Wang, Tongzhen Si, Tao Xu, *Member, IEEE*, Guangqi Li, and Xiaohui Yang, *Member, IEEE*

***Abstract*—Hyperspectral image (HSI) classification is crucial for applications like precision agriculture and urban planning, necessitating the effective integration of spatial and spectral features. However, existing methods often struggle with limited receptive fields, structural damage from data transformation, or high computational complexity, hindering an optimal balance between accuracy and efficiency. To tackle these challenges, we propose HMMamba—a lightweight, accurate, and interpretable heterogeneous multibranch architecture (HMBA). Its core innovation lies in our first proposal of the structured Mamba modeler (SMM), which sequences and reconstructs high-dimensional data through structure-preserving axial compaction and recombination, employing an orientation-separated shared Mamba mechanism to efficiently achieve global dependency modeling. The framework is further augmented by a lightweight anisotropy perceiver (AP) for mid-range spatial structures and a detail enhancer (DE) for local discriminative features. These three complementary branches collectively construct a comprehensive multiscale feature hierarchy. Extensive experiments on five public HSI datasets demonstrate that our method achieves a top overall accuracy (OA) of 98.26%, with merely 0.44 M parameters and an inference speed of 0.01 s, significantly outperforming current convolutional neural network (CNN)-, Transformer-, and Mamba-based models. Experimental results and visualization analyses show that HMMamba excels in structural integrity, classification accuracy, computational efficiency, and interpretability, positioning it as a promising next-generation HSI classifier. The code for this work will be available at https://github.com/IIPGroup-C/HMMamba**

***Index Terms*—Hyperspectral image (HSI) classification, lightweight model, shared Mamba, state-space models (SSMs), structure-preserving axial compaction.**

I. Introduction

AS APPLICATIONS need advancement in various fields, traditional remote sensing images struggle to meet requirements in many scenarios [1]. Hyperspectral imaging records dozens to hundreds of continuous narrow bands per pixel, creating unified spatial–spectral data cubes [2], [3]. This enables capture of discriminative spectral responses unique to ground objects [4], making hyperspectral image (HSI) classification the key bridge from data advantage to practical value [5]. The task relies on joint spatial–spectral information for accurate pixel-level classification [6].

High-accuracy classification is critical for enabling applications from precision agriculture [8], [9] and mineral exploration [10], [11] to environmental monitoring [12], [13], urban planning [14], [15], medical diagnosis [16], [17], and food safety [18], [19]. Despite its potential, HSI classification faces three core challenges: the curse of dimensionality, severe interband redundancy, and poor spatial–spectral synergy [20]. Deep learning has emerged as a key solution [21].

Convolutional neural networks (CNNs) excel in spatial feature extraction via local connectivity and weight sharing [22], [23], [24], [25], but limited receptive fields hinder long-range dependency modeling [26]. Graph convolutional networks (GCNs) address this by modeling pixels as graph nodes to capture irregular contexts [27], [28], [29], yet depend heavily on graph quality and face high computational costs [21]. Transformers advance HSI classification via self-attention, capturing global contextual information [30], [31], [32], [33]. Self-attention connects any pixels, capturing long-range dependencies across the entire image [34], but its quadratic complexity in computation and memory poses severe bottlenecks for practical application [35].

The Mamba architecture [36], based on state-space models (SSMs), offers a promising alternative. Unlike traditional RNNs, SSMs represent systems as continuous-time dynamic models, discretized for sequential data [37], [38], [39], [40]. Mamba introduces selectivity for dynamic parameter adjustment [37], maintaining global receptive fields with linear complexity [41]. However, applying Mamba to HSIs is challenging: it processes 1-D sequences while the HSI is 3-D [42]. "Scan-and-flatten" methods [43], [44], [45] disrupt spatial–spectral topology and produce over-long sequences that undermine Mamba's linear complexity advantage.

Most existing approaches first process spectral and spatial features separately and then fuse them, based on HSI's 3-D structure [41], [44], [45], [46], [47], [48], [49], [50], [51]. This inherently splits the spatial–spectral unity and requires complex designs to reconnect them. Moreover, both mainstream methods have fundamental limitations: patch-based processing disrupts global data structure despite computational feasibility, while whole-image processing suffers from prohibitive computational complexity despite preserving structural integrity.

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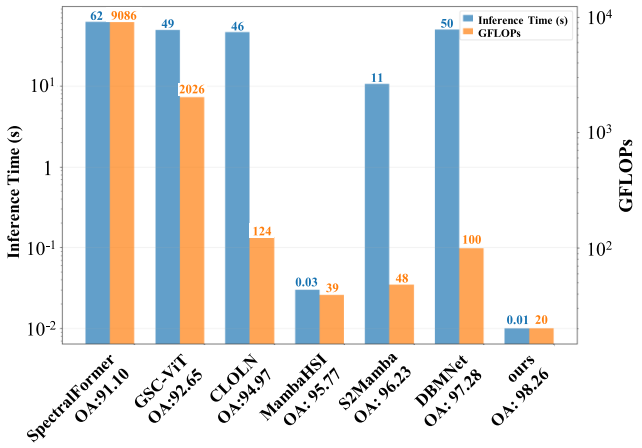


Fig. 1. Performance profile of algorithms on the Pavia University dataset, depicting the interplays between OA (x-axis), inference time (left y-axis), and GFLOPs (right y-axis).

HSI classification has long been constrained by an impossible triangle: existing methods sacrifice either structural integrity, accuracy, or efficiency. We break this limitation with HMMamba, which simultaneously achieves all three objectives through a heterogeneous multibranch architecture (HMBA), overcoming this fundamental tradeoff.

The contribution of this article can mainly be summarized in the following points.

1) A structured Mamba modeler (SMM) is proposed to address accuracy bottlenecks from structural damage. It employs structure-preserving axial compaction for topology-aware sequencing and an orientation-separated shared Mamba for efficient global dependency modeling with linear complexity.
2) A lightweight differentiated architecture is designed to address detail-related challenges in feature extraction, comprising a detail enhancer (DE) that captures fine-grained local features and an anisotropy perceiver (AP) that perceives mid-range structures while bridging local and global semantics.
3) A novel Mamba-based HMBA for HSI classification, named HMMamba, is proposed. By integrating the SMM, DE, and AP, the architecture establishes a global–intermediate–local triple feature extraction pathway. This design achieves efficient spatial–spectral cooperative representation while maintaining linear computational complexity.
4) Extensive experiments on five HSI datasets demonstrate that our method achieves the highest accuracy with notably lower computational cost and faster inference than competitors, as demonstrated in Fig. 1. Additionally, interpretability analyses validate the collaborative effectiveness of our multibranch design.

II. Related Work

The core challenge of HSI classification lies in synergistically utilizing fine-grained spectral discriminability and spatial contextual correlations [1], [2]. This joint spectral–spatial paradigm has driven methodological evolution: from traditional machine learning with handcrafted features, to deep neural networks for end-to-end learning, and now to advanced sequence models for long-sequence modeling [1], [20].

A. ML-Based Methods

Early HSI classification relied on statistical learning and handcrafted features [20], [21]. Support vector machines (SVMs) [52] handled high-dimensional small-sample problems effectively. Random forest (RF) [53] improved stability through ensemble voting. However, these spectral-oriented methods required laborious feature engineering for spatial information incorporation. Techniques like superpixel segmentation [54], extended morphological profiles (EMPs) [55], and multiple kernel learning (MLK) [56] were proposed to address this. Despite notable results, they remained limited by handcrafted feature simplicity and shallow model capacity, struggling to capture complex spatial–spectral interactions in challenging land-cover scenarios.

B. CNN-Based Methods

The advent of deep learning enabled automatic hierarchical feature learning from raw data [21]. CNNs excel at HSI feature extraction due to inductive biases like local connectivity and weight sharing. Hu et al. [22] pioneered 1D-CNN to extract deep spectral features, Lee and Kwon [23] advanced 2D-CNN for capturing spatial textures, and Hamida et al. [24] proposed 3D-CNN for joint spectral–spatial modeling via 3-D convolution. This evolution advanced CNN-based HSI classification from single-modality analysis to collaborative spectral–spatial modeling, propelling further architectural innovations. Roy et al. [57] developed HybridSN, a hybrid 3-D–2-D CNN that balances performance and computational cost. Paoletti et al. [58] incorporated residual connections to mitigate gradient vanishing and improve generalization. Wang et al. [59] utilized dense connections to enhance feature reuse and reduce complexity. Roy et al. [60] incorporated an attention mechanism to dynamically emphasize key spectral bands and spatial regions. However, CNNs' local operations inherently restrict receptive fields, hindering long-range dependency modeling and forming a primary performance bottleneck.

C. GCN-Based Methods

GCNs abstract image pixels as graph nodes and construct edges based on spatial proximity or feature similarity, explicitly modeling complex spatial relationships between pixels through message propagation and aggregation on the graph. Hong et al. [27] pioneered GCNs in HSI classification, demonstrating global context modeling capabilities. Subsequent work optimized graph construction and aggregation: Wan et al. [28] developed a context-aware dynamic GCN that adapts graph structure to input, while Jia et al. [29] proposed GiGCN with hierarchical graphs to model spatial hierarchies. Despite their strengths in non-Euclidean spatial data processing, GCNs remain highly sensitive to graph quality and suffer from high computational and memory costs, restricting large-scale application.

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D. Transformer-Based Methods

Transformers have advanced HSI classification through self-attention, which enables direct pixel interaction to capture long-range dependencies beyond local receptive fields. Building on foundational Vision Transformer works [35], [62], Hong et al. [33] proposed SpectralFormer, pioneering pure Transformers for HSIs by treating patches as token sequences. Subsequent research expanded Transformer applications: Zhou et al. [63] explored multiscan RNN–Transformers for spectral sequencing; Yang et al. [31] introduced bio-inspired center-surround mechanisms; Sun et al. [64] designed Gaussian-weighted dual-branch architectures; and Li et al. [32] developed SPFormer for few-shot learning. To balance global modeling with efficiency, CNN–Transformer hybrids gained focus, such as Roy et al. [60] attention-enhanced ResNet and Zhao et al. [30] GSC–ViT combining grouped convolution with ViT. Despite these advances, Transformers' quadratic complexity remains unresolved, limiting efficiency and scalability on large-scale HSIs.

E. SSM-Based Methods

SSMs [65], [66], particularly structured variants like S4, provide a control-theory-inspired framework for sequence modeling as an alternative to self-attention. The core concept employs a latent state $\boldsymbol{h}(t) \in \mathbb{R}^N$ as system memory, modeling the input–output relationship via linear ODEs

$$\boldsymbol{h}'(t) = \boldsymbol{A}\boldsymbol{h}(t) + \boldsymbol{B}\boldsymbol{x}(t), \quad \boldsymbol{y}(t) = \boldsymbol{C}\boldsymbol{h}(t) + \boldsymbol{D}\boldsymbol{x}(t). \tag{1}$$

SSMs offer dual-mode computation: during training, they parallelize via convolutions; during inference, they recur with constant complexity per step. For discrete sequences, continuous SSMs are discretized

$$\boldsymbol{h}_k = \bar{\boldsymbol{A}}\boldsymbol{h}_{k-1} + \bar{\boldsymbol{B}}\boldsymbol{x}_k, \quad \boldsymbol{y}_k = \boldsymbol{C}\boldsymbol{h}_k \tag{2}$$

where $\bar{\boldsymbol{A}} = \exp(\Delta\boldsymbol{A})$ and $\bar{\boldsymbol{B}} = (\Delta\boldsymbol{A})^{-1}(\exp(\Delta\boldsymbol{A}) - \boldsymbol{I}) \cdot \Delta\boldsymbol{B}$.

While traditional SSMs use static parameters, Mamba introduces selectivity by making $\boldsymbol{B}$, $\boldsymbol{C}$, and $\boldsymbol{\Delta}$ input-dependent

$$\boldsymbol{B}_k = f_B(\boldsymbol{x}_k), \quad \boldsymbol{C}_k = f_C(\boldsymbol{x}_k), \quad \boldsymbol{\Delta}_k = f_\Delta(\boldsymbol{x}_k). \tag{3}$$

This enables dynamic adjustment based on input content, providing contextual awareness with linear complexity, ideal for HSI data.

The rapid introduction of Mamba into HSI classification has sparked significant research interest, with numerous variants addressing different aspects of hyperspectral data characteristics. Li et al. [41] pioneered MambaHSI, demonstrating Mamba's feasibility for HSI by unfolding images into 1-D sequences. This foundational work inspired two main research directions: spectral–spatial modeling enhancement and scanning mechanism improvement. For spectral–spatial modeling, Yao et al. [45] developed SpectralMamba with phase state space to correct spectral distortion, while Wang et al. [46] designed S^{2}Mamba with dual-branch architecture for independent spatial and spectral processing. For scanning mechanisms, Liu et al. [44] introduced VMamba with cross-scanning to mitigate directional bias, inspiring subsequent innovations like Huang et al. [47] bidirectional SS-Mamba and He et al. [48] IGroupSS-Mamba with interval grouping.

Recent advances address 3-D data structure: He et al. [49] constructed 3DSS-Mamba for direct cube modeling, and Zhou et al. [66] designed MIM with nested hierarchy. Gendy and AlMarzouqi [67] proposed GMambaHSI, which introduces multidirectional scanning across four spatial orientations with a modulated group Mamba layer. Liu et al. [68] developed HyperMamba, designing spatial neighborhood adaptive scanning and spectral adaptive enhancement scanning modules. Li et al. [69] proposed HyPyraMamba, incorporating pyramid spectral attention and adaptive expert depthwise convolution for enhanced spectral–spatial representation. Wang et al. [70] then improved this with MambaHSI+, which uses bidirectional state-space modeling and spectral trajectory learning to better preserve spectral continuity.

Hybrid architectures have also emerged, combining Mamba with Transformers [50] and GCNs [51] to leverage complementary strengths.

While promising for long-sequence modeling, existing Mamba-based methods face three challenges. First, flattening 3-D cubes into 1-D sequences or decoupling spatial and spectral features disrupts spatial–spectral topology. Second, unidirectional or cross-scanning mechanisms provide limited contextual perception. Third, direct 3-D cube modeling suffers from excessive sequence length and parameter redundancy, while multimodule designs increase computational overhead. Designing architectures that preserve spatial–spectral structures, capture comprehensive context, and maintain efficiency remains crucial.

III. Methodology

A. Motivation and Overall Idea

The application of Mamba models to HSI classification faces a fundamental challenge: the mismatch between sequential modeling and multidimensional HSI data. Existing methods typically flatten 3-D cubes into 1-D sequences, disrupting spatial–spectral topology and incurring high computational cost. To overcome this, we rethink the pipeline from structural preservation to multiscale representation.

Our approach begins with 3-D axial compression, which projects volumetric data along three dimensions to generate compact, topology-preserving short sequences. These are processed through directionally separated, parameter-shared Mamba blocks to maintain directional specificity efficiently.

We extend the structured sequence modeling approach to multiscale feature extraction by developing complementary lightweight pathways that collaborate with the sequence modeler. This heterogeneous architecture ensures comprehensive representation learning across spatial and spectral dimensions while maintaining computational efficiency.

B. Overview

This article introduces HMMamba, an HMBA for HSI classification. As shown in Fig. 2 and Algorithm 1, this end-to-end framework processes HSI data through four stages.

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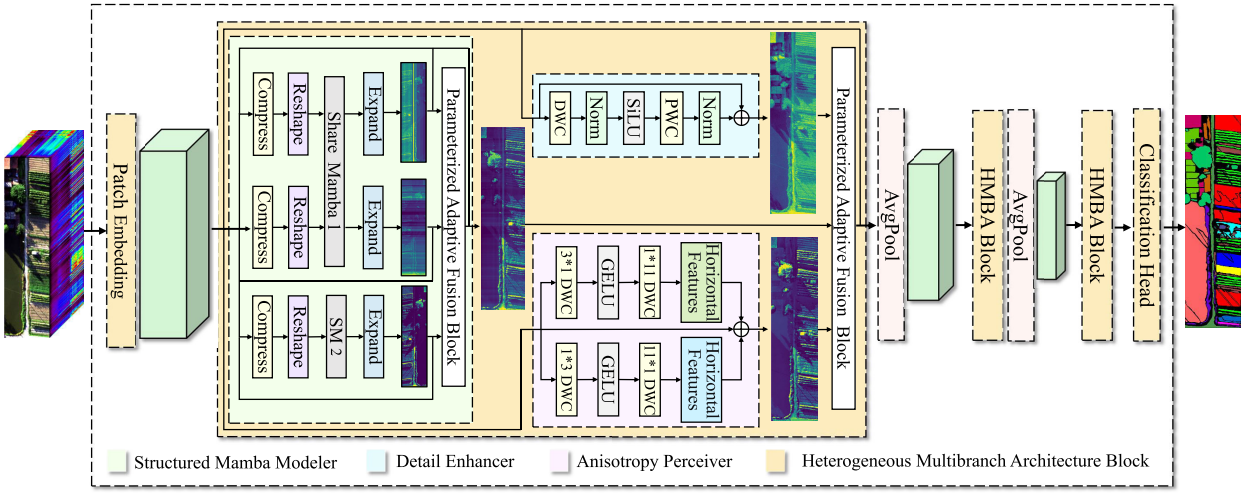


Fig. 2. Overview of HMMamba. The HMBA Block comprises SMM for global sequence modeling with axial compression, DE for local feature refinement, and AP for mid-range structure modeling. The outputs are fused for pixel-wise classification.

Algorithm 1 HMMamba

Input: HSI data $\boldsymbol{I} \in \mathbb{R}^{H\times W\times C}$
Output: Classification map $\hat{\boldsymbol{Y}}$
Feature Embedding:
$\boldsymbol{E}_0 = \mathrm{SiLU}(\mathrm{GN}(\mathrm{Conv}_{1\times1}(\boldsymbol{I})))$
Multiscale Feature Pyramid:
for $i = 1$ to N **do**
  HMBA Module:
  $\boldsymbol{H}_\mathrm{g} = \mathrm{SMM}(\boldsymbol{E}_{i-1})$ // Global dependency modeling
  $\boldsymbol{H}_\mathrm{m} = \mathrm{AP}(\boldsymbol{E}_{i-1})$ // Mid-range context
  $\boldsymbol{H}_\mathrm{l} = \mathrm{DE}(\boldsymbol{E}_{i-1})$ // Local details
  $\boldsymbol{H} = \delta_g\boldsymbol{H}_\mathrm{g} + \delta_m\boldsymbol{H}_\mathrm{m} + \delta_l\boldsymbol{H}_\mathrm{l} + \boldsymbol{E}_{i-1}$
  Spatial Downsampling:
  $\boldsymbol{E}_i = \mathrm{Downsample}(\boldsymbol{H})$
end for
Classification Head:
$\hat{\boldsymbol{Y}} = \mathrm{Conv}(\mathrm{SiLU}(\mathrm{GN}(\mathrm{Conv}(\boldsymbol{E}_N))))$
Return $\hat{\boldsymbol{Y}}$

First, feature embedding projects $\boldsymbol{I} \in \mathbb{R}^{H\times W\times C}$ to $\boldsymbol{E} \in \mathbb{R}^{H\times W\times D}$ via 1×1 convolution, GN, and SiLU

$$\boldsymbol{E} = \mathrm{Embedding}(\boldsymbol{I}). \tag{4}$$

The core HMBA module employs three parallel branches—SMM, AP, and DE—for global, mid-range, and local feature extraction, fused adaptively with residual connections

$$\boldsymbol{H} = \mathrm{HMBA}(\boldsymbol{E}). \tag{5}$$

Alternating HMBA blocks with downsampling layers construct a multiscale feature pyramid. Finally, a lightweight classification head produces pixel-wise predictions

$$\hat{\boldsymbol{Y}} = \mathrm{ClsHead}(\boldsymbol{H}). \tag{6}$$

The primary innovation is the heterogeneous multibranch framework. The feature embedding and classification head follow MambaHSI [41] to focus on core contributions.

C. Global Dependency Modeling Branch—SMM

The global dependency modeling branch addresses preserving HSI structural integrity while maintaining efficiency. Conventional scan-and-flatten methods map $\boldsymbol{I} \in \mathbb{R}^{H\times W\times C}$ to $\boldsymbol{S} \in \mathbb{R}^{L}$ ($L = H\times W\times C$) by concatenating rows or scanning in various orders. Such operations inevitably place spatially nonadjacent pixels (e.g., the end of one row and the beginning of the next) into consecutive sequence positions, creating artificial spatial correlations while disrupting natural adjacency relationships. Consequently, the intrinsic spatial–spectral topology is distorted.

In contrast, our SMM preserves structural integrity through three key designs. First, instead of concatenating spatially disconnected pixels, we perform 3-D axial compression that compresses along one dimension at a time via pooling operations. Specifically, width–axis compression reduces the width dimension to size one while preserving the complete height–spectral relationship (H–C); height–axis compression preserves width–spectral correlation (W–C); spectral–axis compression preserves the full 2-D spatial layout (H–W). No concatenation across disconnected regions is involved. Second, we construct multiperspective representations through H–W, H–C, and W–C pairwise combinations, treating hyperspectral data as an indivisible unity rather than decoupling spatial and spectral features. Third, the three complementary projections are adaptively fused to recover the complete 3-D spatial–spectral context. This design reduces complexity while enabling efficient long-sequence modeling. Fig. 3 and Algorithm 2 detail the SMM workflow and compression process.

We define $\mathrm{BiLinearPool}(\boldsymbol{X},\cdot)$ as the parallel application of adaptive average and max-pooling

$$\mathrm{BiLinearPool}(\boldsymbol{X},\cdot) = \left\{\begin{matrix}\mathrm{AdaptiveAvgPool2d}(\boldsymbol{X},\cdot)\\ \mathrm{AdaptiveMaxPool2d}(\boldsymbol{X},\cdot)\end{matrix}\right\}. \tag{7}$$

First, multiaxial compression performs bilinear pooling along three orthogonal dimensions on $\boldsymbol{X} \in \mathbb{R}^{B\times C\times H\times W}$.

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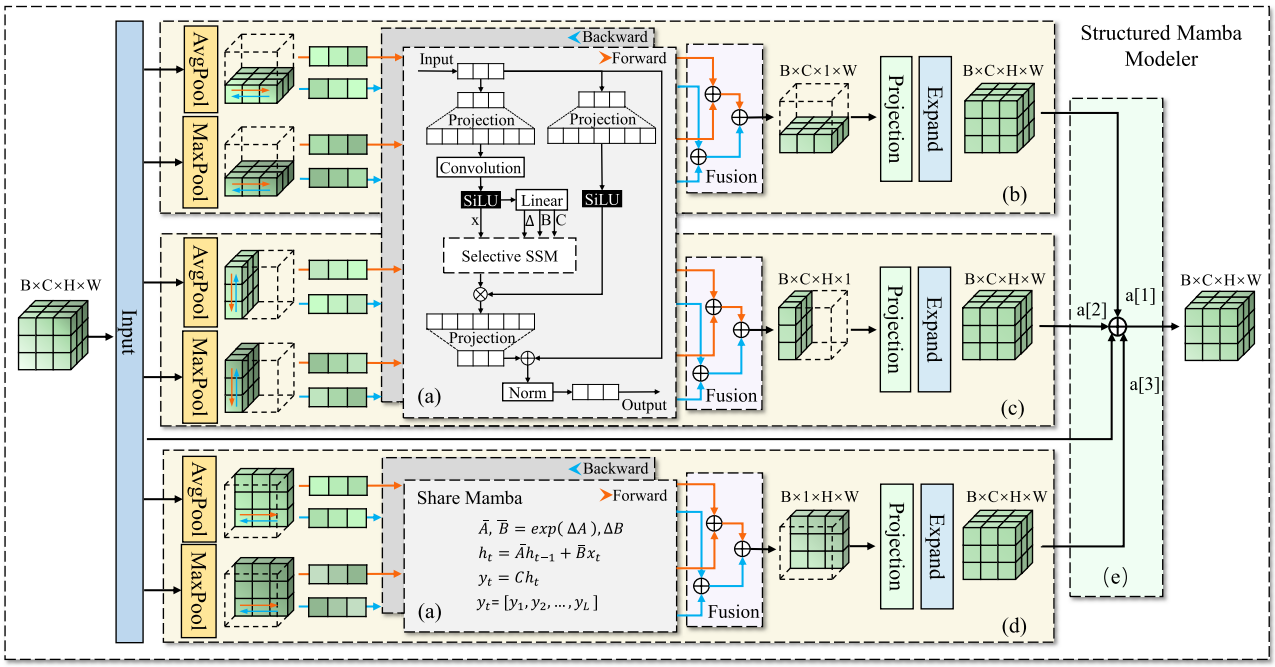


Fig. 3. SMM design. The diagram details the 3-D axial compression process with (a) share Mamba, (b) width–axis compression, (c) height–axis compression, (d) spectral–axis compression and (e) parameterized adaptive fusion.

Each $\boldsymbol{X}^{(\cdot)}$ contains two feature maps from average and max-pooling: $\boldsymbol{X}^w = \{\boldsymbol{X}^w_{\text{avg}}, \boldsymbol{X}^w_{\text{max}}\} \in \mathbb{R}^{B\times C\times H\times 1}$ preserves height–spectral relationships, $\boldsymbol{X}^h = \{\boldsymbol{X}^h_{\text{avg}}, \boldsymbol{X}^h_{\text{max}}\} \in \mathbb{R}^{B\times C\times 1\times W}$ maintains width–spectral correlations, and $\boldsymbol{X}^c = \{\boldsymbol{X}^c_{\text{avg}}, \boldsymbol{X}^c_{\text{max}}\} \in \mathbb{R}^{B\times H\times W\times 1}$ preserves spatial structure.

As illustrated in Fig. 4, the six compressed feature maps are reshaped into 8 feature sequences $\boldsymbol{S}$ for Mamba processing.

This process yields two sequence types: spatial–axis $(\boldsymbol{S}^w, \boldsymbol{S}^h)$ and spectral–axis $(\boldsymbol{S}^c_w, \boldsymbol{S}^c_h)$. Spatial–axis sequences preserve spectral resolution while compressing spatial dimensions, forming a position-spectrum flow where each element integrates all spectral information at a spatial location. Spectral–axis sequences aggregate spatial information while compressing spectral dimensions, forming a position-position flow where each element represents a global spatial summary per channel, modeling interpixel correlations.

Subsequently, Mamba processing employs a dual-path parameter-sharing mechanism where spatial–spectral sequences $\boldsymbol{S}^h$ and $\boldsymbol{S}^w$ share one Mamba module, while global spatial sequences $\boldsymbol{S}^c_w$ and $\boldsymbol{S}^c_h$ share another (Fig. 5). Each shared Mamba generates input-specific parameters

$$\begin{aligned}\boldsymbol{B}^i &= \text{Linear}_B\left(\boldsymbol{S}^i\right)\\ \boldsymbol{C}^i &= \text{Linear}_C\left(\boldsymbol{S}^i\right)\\ \boldsymbol{\Delta}^i &= \text{SoftPlus}\left(\text{Linear}_\Delta\left(\boldsymbol{S}^i\right)\right) + \text{bias}_\Delta\end{aligned} \tag{8}$$

where bias_Δ is a learnable bias, and SoftPlus ensures positive discretization step sizes. This parameterization enables dynamic state updates and output generation based on input content.

Discretization yields $\bar{\boldsymbol{A}}^i, \bar{\boldsymbol{B}}^i = \exp(\Delta^i \boldsymbol{A}), \Delta^i \boldsymbol{B}^i$, where Δ^i balances current input influence versus historical state dependence. Selective scanning operates in dual modes: parallel convolution with global kernel $\boldsymbol{K}$ produces hidden states $\boldsymbol{HS} = \text{SSM_Conv}(\boldsymbol{S}, \boldsymbol{K})$, while recurrent computation follows $\boldsymbol{h}(t) = \bar{\boldsymbol{A}}(t)\boldsymbol{h}(t-1) + \bar{\boldsymbol{B}}(t)\boldsymbol{S}(t)$ for inference efficiency.

The dynamic parameter $\boldsymbol{C}$ is used to generate outputs from the hidden states, with residual connections added to ensure stability

$$\boldsymbol{M} = (\boldsymbol{C} \times \boldsymbol{HS}) + \text{Linear_Out}(\boldsymbol{S}). \tag{9}$$

To overcome unidirectional Mamba's causal limitation, we apply bidirectional processing. For width–axis sequences

$$\begin{aligned}\boldsymbol{M}^w_{\text{for}} &= \text{ShareMamba}_1\left(\boldsymbol{S}^w\right)\\ \boldsymbol{M}^w_{\text{back}} &= \text{Reverse}\left(\text{ShareMamba}_1\left(\text{Reverse}\left(\boldsymbol{S}^w\right)\right)\right).\end{aligned} \tag{10}$$

Bidirectional features are averaged

$$\boldsymbol{M} = \left(\boldsymbol{M}_{\text{for}} + \boldsymbol{M}_{\text{back}}\right)/2. \tag{11}$$

The sequences are then reshaped to their original spatial dimensions through dimension reconstruction

$$\begin{aligned}\boldsymbol{M}_R &= \text{Rearrange}\left(\boldsymbol{M}, \text{b h c} \to \text{b c h 1}\right)\\ \boldsymbol{M}_{out} &= \text{Expand}\left(\boldsymbol{Y}, [-1, -1, -1, W]\right)\end{aligned} \tag{12}$$

and enhanced through a shared nonlinear projection (GroupNorm + SiLU).

Finally, outputs from average and max-pooling paths are adaptively fused for each axial direction

$$\boldsymbol{F} = \alpha_0 \cdot \boldsymbol{M}^{\text{out}}_{\text{max}} + \alpha_1 \cdot \boldsymbol{M}^{\text{out}}_{\text{avg}} + \boldsymbol{X} \tag{13}$$

where α_0 and α_1 are normalized fusion coefficients, and $\boldsymbol{X}$ is the residual connection. This yields the final outputs $\boldsymbol{F}^w$, $\boldsymbol{F}^h$, and $\boldsymbol{F}^c$ for the width, height, and spectral branches, respectively.

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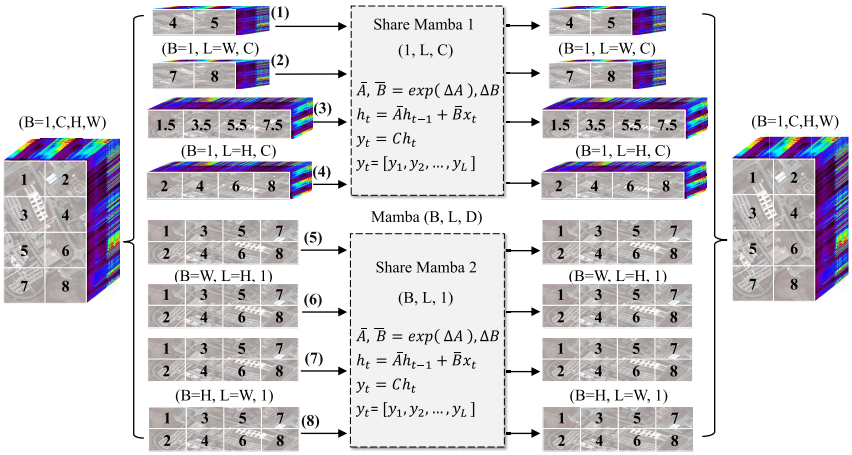


Fig. 4. Multiaxial sequence processing framework (forward direction only; backward direction is processed similarly by flipping each sequence). The input HSI cube $\boldsymbol{X} \in \mathbb{R}^{H\times W\times C}$ is processed into eight sequences formatted as (B, L, D) for Mamba requirements. (1) and (2) Height–axis compression: $\boldsymbol{S}^h_{\text{avg}}$ and $\boldsymbol{S}^h_{\max}$ with shape $(1, W, C)$. (3) and (4) Width–axis compression: $\boldsymbol{S}^w_{\text{avg}}$ and $\boldsymbol{S}^w_{\max}$ with shape $(1, H, C)$. (5) and (6) Spectral–axis compression (W as batch): $\boldsymbol{S}^c_{w,\text{avg}}$ and $\boldsymbol{S}^c_{w,\max}$ with shape $(W, H, 1)$. (7) and (8) Spectral–axis compression (H as batch): $\boldsymbol{S}^c_{h,\text{avg}}$ and $\boldsymbol{S}^c_{h,\max}$ with shape $(H, W, 1)$. All processed sequences are reshaped back to reconstruct the output HSI feature map.

Algorithm 2 SMM

Input: Feature map $\boldsymbol{X} \in \mathbb{R}^{B\times C\times H\times W}$
Output: Global feature map $\boldsymbol{H}_{\text{g}}$
Multiaxial Compression:
$\boldsymbol{X}^w = \text{BiLinearPool}(\boldsymbol{X}, (H, 1))$
$\boldsymbol{X}^h = \text{BiLinearPool}(\boldsymbol{X}, (1, W))$
$\boldsymbol{X}^c = \text{BiLinearPool}(\text{Permute}(\boldsymbol{X}, (0, 2, 3, 1)), (W, 1))$
Sequence Generation:
$\boldsymbol{S}^w = \text{Rearrange}(\boldsymbol{X}^w, b\ c\ h\ 1 \to b\ h\ c)$
$\boldsymbol{S}^h = \text{Rearrange}(\boldsymbol{X}^h, b\ c\ 1\ w \to b\ w\ c)$
$\boldsymbol{S}^c_w = \text{Rearrange}(\boldsymbol{X}^c, b\ h\ w\ 1 \to bh\ w\ 1)$
$\boldsymbol{S}^c_h = \text{Rearrange}(\boldsymbol{X}^c, b\ h\ w\ 1 \to bw\ h\ 1)$
Shared Mamba with Bidirectional Processing:
// 4 directions share Mamba SM_1
for each $\boldsymbol{S}$ in $\{\boldsymbol{S}_w, \boldsymbol{S}_h\}$ **do**
  $\boldsymbol{M} = (\text{SM}_1(\boldsymbol{S}) + \text{Flip}(\text{SM}_1(\text{Flip}(\boldsymbol{S}))))/2$
end for
// 4 directions share Mamba SM_2
for each $\boldsymbol{S}$ in $\{\boldsymbol{S}^c_w, \boldsymbol{S}^c_h\}$ **do**
  $\boldsymbol{M} = (\text{SM}_2(\boldsymbol{S}) + \text{Flip}(\text{SM}_2(\text{Flip}(\boldsymbol{S}))))/2$
end for
Dimension Reconstruction:
$\boldsymbol{F}_w = \text{Expand}(\text{Rearrange}(\boldsymbol{M}_w, b\ h\ c \to b\ c\ h\ 1))$
$\boldsymbol{F}_h = \text{Expand}(\text{Rearrange}(\boldsymbol{M}_h, b\ w\ c \to b\ c\ 1\ w))$
$\boldsymbol{F}_c = \text{Rearrange}(\boldsymbol{M}_c, b\ h\ w \to b\ c\ h\ w)$
$\boldsymbol{F}_w, \boldsymbol{F}_h, \boldsymbol{F}_c = \text{Proj}(\boldsymbol{F}_w, \boldsymbol{F}_h, \boldsymbol{F}_c)$ // GroupNorm + SiLU
Parameterized Adaptive Fusion:
$\boldsymbol{H}_g = \beta_w \cdot \boldsymbol{F}_w + \beta_h \cdot \boldsymbol{F}_h + \beta_c \cdot \boldsymbol{F}_c + X$
return: $\boldsymbol{H}_g$

Subsequently, tri-axial feature fusion integrates the enhanced representations through learnable weights

$$\boldsymbol{H}_g = \beta_w \cdot \boldsymbol{F}^w + \beta_h \cdot \boldsymbol{F}^h + \beta_c \cdot \boldsymbol{F}^c + \boldsymbol{X} \tag{14}$$

where β_w, β_h, and β_c are normalized fusion coefficients that dynamically balance axial contributions based on input characteristics.

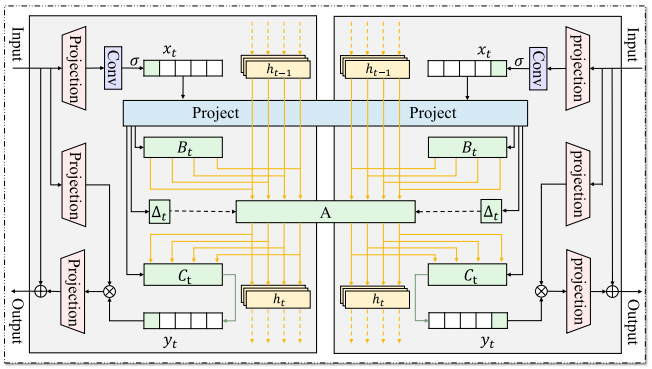


Fig. 5. Shared Mamba processing pipeline. The diagram illustrates the parameter sharing mechanism between two homogeneous sequences. The left half shows the processing flow of the first sequence, while the right half shows the parallel flow of the second sequence, demonstrating parameter reuse across sequences.

The framework processes 16 sequences in total through two shared Mamba modules

$$\begin{aligned}
&\text{Group 1 (spatial−axis):}\\
&\left\{\boldsymbol{S}^{\text{avg}}_{w,for}, \boldsymbol{S}^{\text{avg}}_{w,back}, \boldsymbol{S}^{\max}_{w,for}, \boldsymbol{S}^{\max}_{w,back},\right.\\
&\left.\boldsymbol{S}^{\text{avg}}_{h,for}, \boldsymbol{S}^{\text{avg}}_{h,back}, \boldsymbol{S}^{\max}_{h,for}, \boldsymbol{S}^{\max}_{h,back}\right\} \xrightarrow{\text{SM}_1} \{\cdot\}\\
&\text{Group 2 (spectral−axis):}\\
&\left\{\boldsymbol{S}^{h,\text{avg}}_{c,for}, \boldsymbol{S}^{h,\text{avg}}_{c,back}, \boldsymbol{S}^{h,\max}_{c,for}, \boldsymbol{S}^{h,\max}_{c,back},\right.\\
&\left.\boldsymbol{S}^{w,\text{avg}}_{c,for}, \boldsymbol{S}^{w,\text{avg}}_{c,back}, \boldsymbol{S}^{w,\max}_{c,for}, \boldsymbol{S}^{w,\max}_{c,back}\right\} \xrightarrow{\text{SM}_2} \{\cdot\}
\end{aligned} \tag{15}$$

where SM_1 and SM_2 denote the two shared Mamba modules. This design achieves an $8\times$ parameter reduction compared to using 16 independent sequence-level Mambas. It also achieves a $2\times$ reduction compared to using four direction-level Mambas ($\boldsymbol{S}^h$, $\boldsymbol{S}^w$, $\boldsymbol{S}^c_w$, and $\boldsymbol{S}^c_h$). Linear complexity is maintained.

Compared to existing methods, SMM offers three key advantages: 3-D axial compression preserves data structure,

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parameter sharing reduces model complexity, and bidirectional processing enhances contextual modeling. These features collectively enable effective long-range dependency capture in HSI data while maintaining computational efficiency.

D. Mid-Range Context Branch—AP

The mid-range context capture branch introduces the AP to model medium-range spatial dependencies efficiently. AP employs separated convolutional groups that decompose 2-D convolution into cascaded 1-D operations, maintaining representational capacity while reducing complexity.

As shown in Fig. 2, AP consists of three components. The horizontal module captures width–dimension context through 3×1 and 1×11 depthwise convolutions, while the vertical module employs symmetric operations with 1×3 and 11×1 kernels

$$\begin{aligned} \boldsymbol{F}_h &= \mathrm{DWConv}_{1\times 11}\left(\mathrm{GELU}\left(\mathrm{DWConv}_{3\times 1}\left(\boldsymbol{X}\right)\right)\right) \\ \boldsymbol{F}_v &= \mathrm{DWConv}_{11\times 1}\left(\mathrm{GELU}\left(\mathrm{DWConv}_{1\times 3}\left(\boldsymbol{X}\right)\right)\right). \end{aligned} \tag{16}$$

Features are adaptively fused as

$$\boldsymbol{H}_m = \gamma_h \cdot \boldsymbol{F}_h + \gamma_v \cdot \boldsymbol{F}_v + \boldsymbol{X} \tag{17}$$

where γ_h and γ_v are learned fusion coefficients.

This design achieves significant complexity reduction. Compared to standard 3×11 2-D convolution ($O = 33BC^2HW$), our separated depthwise convolutions require only $O = 14BCHW$ operations, achieving 2.36 C speedup (approximately $300\times$ when $C = 128$).

AP offers three advantages: efficient large receptive field via separated convolutions, anisotropic pattern capture through directional modeling, and dynamic feature importance adjustment. It effectively processes directional features like farmland furrows and urban grids for mesoscale land-cover recognition.

E. Local Details Branch—DE

The local details branch employs the DE for efficient pixel-level feature extraction. DE uses depthwise separable convolution to capture fine-grained textures and edges with minimal computational overhead.

Features are first processed through 3×3 depthwise convolution with BatchNorm and SiLU activation for spatial feature extraction, followed by 1×1 pointwise convolution with BatchNorm for cross-channel fusion. A residual connection adds the processed features to the original input

$$\boldsymbol{F}_{\mathrm{DW}} = \mathrm{SiLU}\left(\mathrm{BatchNorm}\left(\mathrm{DWConv}_{3\times 3}\left(\boldsymbol{X}\right)\right)\right) \tag{18}$$

$$\boldsymbol{H}_l = \mathrm{BatchNorm}\left(\mathrm{Conv}_{1\times 1}\left(\boldsymbol{F}_{\mathrm{DW}}\right)\right) + \boldsymbol{X}. \tag{19}$$

This design achieves significant complexity reduction from $O(BC^2HWk^2)$ to $O(BCHW(k^2 + C))$, providing $8.4\times$ speedup when $C = 128$ while maintaining effective local feature extraction.

DE offers three advantages: lightweight design with minimal parameters, effective local texture and edge preservation, and residual connections ensuring gradient flow. It specializes in capturing essential spatial details like building contours and individual trees for precise boundary localization.

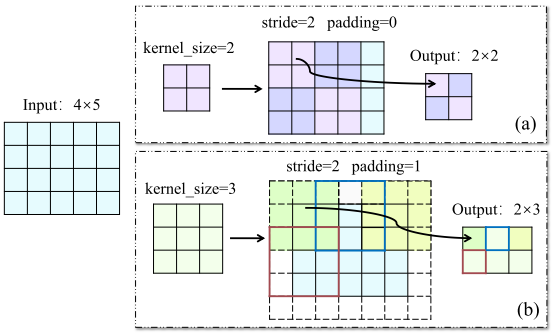


Fig. 6. Comparison of downsampling strategies on a 4×5 input example. (a) MambaHSI downsampling (kernel = 2, stride = 2, padding = 0): odd dimension reduces $5 \rightarrow 2$, losing boundary information. (b) Our downsampling (kernel = 3, stride = 2, padding = 1): odd dimension reduces $5 \rightarrow 3$, preserving all boundaries. Both methods produce identical results for even dimensions $4 \rightarrow 2$.

Finally, the HMBA integrates features through adaptive fusion

$$\boldsymbol{H} = \left(\delta_g \cdot \boldsymbol{H}_g + \delta_m \cdot \boldsymbol{H}_m + \delta_l \cdot \boldsymbol{H}_l\right) + \boldsymbol{X} \tag{20}$$

where δ_g, δ_m, and δ_l are learnable coefficients. This enables dynamic scale preference adjustment based on input characteristics, achieving intelligent feature routing. The output $\boldsymbol{Z}$ integrates local textures, mid-range context, and global dependencies into a discriminative spatial–spectral representation for classification.

F. Multiscale Feature Pyramid

To enhance multiscale recognition capability, we construct a feature pyramid by alternately stacking heterogeneous multibranch modules with spatial downsampling layers. Three scales are implemented: high-resolution shallow features, medium-range regional semantics after first downsampling, and scene-level understanding after further downsampling.

Downsampling uses convolution (kernel = 3, stride = 2, padding = 1). Compared to MambaHSI's aggressive downsampling (kernel = 2, stride = 2, padding = 0), our design better preserves boundary information (Fig. 6). The difference is quantified as

$$\boldsymbol{H}_{\mathrm{out}}^{\mathrm{MambaHSI}} = \left\lfloor \frac{\boldsymbol{H}_{\mathrm{in}} - 2}{2} \right\rfloor + 1, \quad \boldsymbol{H}_{\mathrm{out}}^{\mathrm{Ours}} = \left\lfloor \frac{\boldsymbol{H}_{\mathrm{in}} - 1}{2} \right\rfloor + 1 \tag{21}$$

where $\lfloor \cdot \rfloor$ denotes the floor operation. For odd $\boldsymbol{H}_{\mathrm{in}}$, our method preserves boundaries while MambaHSI loses information.

In summary, our downsampling design ensures predictable behavior for odd-sized inputs. As an overlapping convolution operation, it performs effective feature extraction with larger receptive fields while reducing resolution, rather than simply discarding input information. This approach achieves an optimal balance between information preservation and computational efficiency.

IV. Experiments

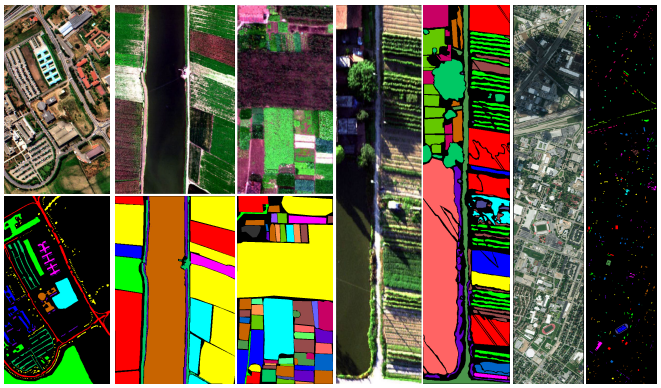
This section presents a comprehensive experimental evaluation of our proposed method.

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TABLE I

DATASET DISTRIBUTION BY CATEGORY FOR PAVIA UNIVERSITY, LONGKOU, HONGHU, HANCHUAN, AND HOUSTON

PaviaU						LongKou						HongHu						HanChuan						Houston					
Color	Category	Train	Val	Test	Total	Color	Category	Train	Val	Test	Total	Color	Category	Train	Val	Test	Total	Color	Category	Train	Val	Test	Total	Color	Category	Train	Val	Test	Total
	Asphalt	30	10	6591	6631		Corn	30	10	34471	34511		Red roof	30	10	14001	14041		Strawberry	30	10	44695	55735		Healthy Grass	30	10	1211	1251
	Meadows	30	10	18609	18649		Cotton	30	10	8334	8374		Road	30	10	3472	3512		Cowpea	30	10	22713	22753		Stressed Grass	30	10	1214	1254
	Gravel	30	10	2059	2099		Sesame	30	10	2991	3031		Bare soil	30	10	21781	21821		Soybean	30	10	10247	10287		Synthetic Grass	30	10	657	697
	Trees	30	10	3024	3064		Broad-leaf soybean	30	10	63172	63212		Cotton	30	10	163245	163285		Sorghum	30	10	5313	5353		Tree	30	10	1204	1244
	Metal sheets	30	10	1305	1345		Narrow-leaf soybean	30	10	4111	4151		Cotton firewood	30	10	6178	6218		Water spinach	30	10	1160	1200		Soil	30	10	1202	1242
	Bare soil	30	10	4989	5029		Rice	30	10	11814	11854		Rape	30	10	44517	44557		Watermelon	30	10	4493	4533		Water	30	10	285	325
	Bitumen	30	10	1290	1330		Water	30	10	67016	67056		Chinese cabbage	30	10	24063	24103		Greens	30	10	5863	5903		Residential	30	10	1228	1268
	Bricks	30	10	3642	3682		Roads and houses	30	10	7084	7124		Pakchoi	30	10	4014	4054		Trees	30	10	17938	17978		Commercial	30	10	1204	1244
	Shadows	30	10	907	947		Mixed weed	30	10	5189	5229		Cabbage	30	10	10779	10819		Grass	30	10	9429	9469		Road	30	10	1212	1252
													Tuber mustard	30	10	12354	12394		Red roof	30	10	10476	10516		Highway	30	10	1187	1227
													Brassica parachinensis	30	10	10975	11015		Gray roof	30	10	16871	16911		Railway	30	10	1195	1235
													Brassica chinensis	30	10	8914	8954		Plastic	30	10	3639	3679		Parking Lot 1	30	10	1193	1233
													Small Brassica chinensis	30	10	22467	22507		Bare soil	30	10	9076	9116		Parking Lot 2	30	10	429	469
													Lactuce sativa	30	10	7316	7356		Road	30	10	18520	18560		Tennis Court	30	10	388	428
													Celtuce	30	10	962	1002		Bright object	30	10	1096	1136		Running Track	30	10	620	660
													Film covered lettuce	30	10	7222	7262		Water	30	10	75361	75401						
													Romaine lettuce	30	10	2970	3010												
													Carrot	30	10	3177	3217												
													White radish	30	10	8672	8712												
													Garlic sprout	30	10	3446	3486												
													Broad bean	30	10	1288	1328												
													Tree	30	10	4000	4040												
Total		270	90	42416	42776	Total		270	90	204182	204542	Total		660	220	385813	386693	Total		480	160	256890	257530	Total		450	150	14429	15029



(a) (b) (c) (d) (e)

Fig. 7. Experimental HSI false-color images and ground truth across five regions. (a) PaviaU. (b) LongKou. (c) HongHu. (d) HanChuan. (e) Houston.

A. Experimental Setup

Datasets: To validate the proposed method, we conducted experiments on five public hyperspectral datasets: Pavia University, WHU-Hi-LongKou, WHU-Hi-HongHu, WHU-Hi-HanChuan [71], and Houston. These datasets represent diverse remote sensing scenarios with varying spatial resolutions and land-cover complexities, effectively evaluating model robustness. Fig. 7 shows their false-color composites and ground truth labels, while Table I summarizes the data characteristics and splits.

1) PaviaU: Acquired by ROSIS over Pavia University, Italy (103 bands, 610 × 340 pixels, 1.3 m resolution). Covers 9 urban categories (42 776 samples). Main challenges are intra-class spectral variations in urban materials and mixed pixels.

2) LongKou: Acquired over Longkou, Hubei, comprising 270 spectral bands (550 × 400 pixels, 0.463 m resolution). Contains nine categories, including six crop types and three background classes, totaling 204 542 samples. Challenges include fine-grained discrimination of morphologically similar crops and distinguishing crops from background vegetation.

3) HongHu: Acquired over Honghu Lake, Hubei, with 270 bands (940 × 475 pixels, 0.043 m resolution). Contains 22 vegetable categories (386 693 samples). The core challenge is discriminating spectrally similar crops with significant intra-class variations.

4) HanChuan: Captured over Hanchuan, Hubei, comprising 274 spectral bands (1217 × 303 pixels, 0.109 m resolution). Covers 16 land-cover categories with 257 530 samples. Challenges include distinguishing natural vegetation from artificial structures in agro-urban mixed scenarios with high heterogeneity.

5) Houston: Acquired by CASI-1500 over Houston, USA (144 bands, 349 × 1905 pixels, 2.5 m resolution). Contains 15 categories (15 029 samples). Challenges include illumination variations, elongated distribution, and class imbalance.

a) Evaluation metrics: To ensure fair and rigorous evaluation, we adopted three widely recognized metrics in hyperspectral image classification: overall accuracy (OA), average accuracy (AA), and Kappa coefficient. All experiments were repeated ten times to mitigate bias from random training sample selection. Larger mean values and smaller standard deviations of these metrics indicate better model performance.

b) Implementation details: The framework is implemented in PyTorch with ten independent runs using distinct random seeds. Data partitioning follows Table I, where for each class, 30 samples are randomly selected as the training set, 10 samples as the validation set, and the remaining samples as the test set. Whole-image processing sets the batch size to 1 for complete prediction map generation. The Adam optimizer is used with learning rate 3×10^{-4}, hidden dimension 128, 3 stacked HMBA blocks, and Mamba state dimension 16. All runs were on the same hardware: an NVIDIA RTX 3090 GPU, an Intel Xeon Gold 6226 R CPU, and 256 GB RAM.

c) Comparison methods: A comprehensive evaluation was conducted by comparing the proposed method with several representative HSI classifiers based on different advanced paradigms:

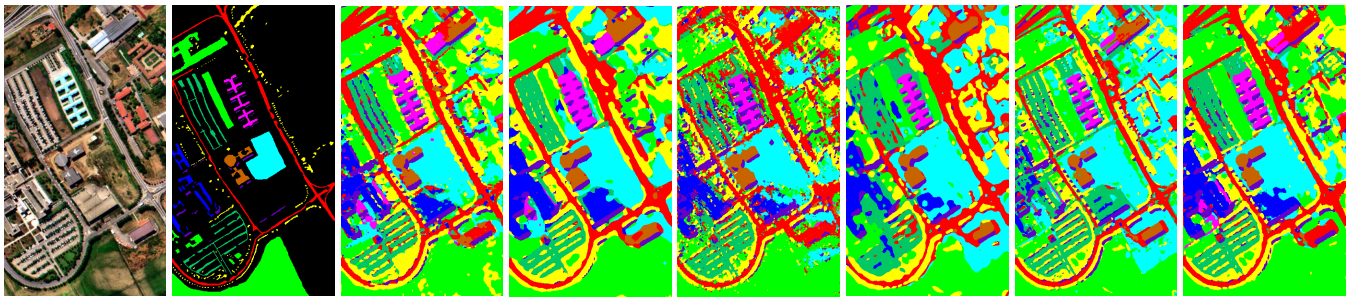
1) *CNN-Based Methods:* CLOLN [25] [TGRS 2024].
2) *GCN-Based Methods:* GiGCN [27] [TNNLS 2022].
3) *Transformer-Based Methods:* GSC–ViT [30] [TGRS 2024].

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TABLE II

QUANTITATIVE RESULTS (ACC% ± STD%) OF PAVIAU. BEST IN BOLD AND SECOND WITH UNDERLINE. IF TWO VALUES ARE IDENTICAL, THE ONE WITH THE SMALLER STD IS CONSIDERED THE BETTER. THESE NOTES ARE THE SAME AS OTHERS

Method	GiGCN	CLONL	GSC-VIT	MambaHSI	IGroupSS	S2Mamba	MambaHSI+	GMambaHSI	HypraMamba	HSIMFormer	DBMGNet	Ours
Asphalt	90.36±1.73	**98.27±1.00**	97.10±1.64	94.42±1.76	93.41+3.24	98.10±0.23	97.31±2.60	94.48±2.45	88.15±3.48	91.04±3.10	96.69±0.38	95.80±2.72
Meadows	98.86±0.78	**98.98±0.76**	97.68±1.32	96.50±1.90	94.86+2.53	93.94±0.48	97.94±1.42	96.71±1.85	93.35±4.59	94.19±2.73	97.82±0.81	98.62±1.62
Gravel	91.66±6.33	87.83±7.44	86.46±7.42	93.03±5.55	92.58+3.91	**99.60±0.34**	98.56±1.83	98.38±2.58	95.11±3.49	91.64±3.39	96.00±1.06	97.21±1.25
Trees	81.34±10.11	97.80±1.46	91.83±9.94	89.96±3.65	93.11+3.42	**98.24±0.59**	94.96±1.46	95.68±1.42	92.66±2.08	92.42±3.20	97.56±0.48	97.47±0.54
Metal sheets	91.15±1.31	99.68±0.59	99.45±0.53	99.99±0.02	99.23+0.45	99.46±0.38	99.87±0.22	99.90±0.15	99.96±0.06	99.73±0.26	**100.00±0.00**	99.90±0.13
Bare soil	98.87±1.90	85.27±7.50	79.65±9.24	98.53±1.00	95.76+1.88	97.26±1.38	**99.96±0.07**	99.93±0.17	98.58±2.38	98.01±1.95	96.78±1.06	99.81±0.55
Bitumen	99.28±0.61	85.92±7.00	89.63±10.10	96.47±3.13	96.43+2.16	99.81±0.12	99.66±0.50	99.66±0.50	**99.90±0.12**	99.81±0.20	99.62±0.29	99.43±0.38
Bricks	87.68±7.46	90.93±4.12	89.50±1.70	94.44±2.85	94.25+2.27	96.35±0.93	97.66±1.92	97.85±0.93	92.41±5.97	90.98±3.26	94.31±1.80	**98.50±1.57**
Shadows	97.21±3.36	97.12±0.33	98.71±1.10	99.37±0.59	97.89+0.76	99.36±0.63	98.36±1.26	98.19±1.33	97.93±1.31	96.21±1.51	**99.89±0.10**	99.47±0.59
OA(%)	94.47±0.97	94.97±1.49	92.65±2.30	95.77±0.89	94.18±1.12	96.23±0.21	97.99±0.74	97.22±0.80	93.61±2.02	94.01±1.35	97.28±0.41	**98.24±0.85**
AA(%)	92.93±1.07	93.53±1.53	91.89±1.73	95.86±1.12	95.98±0.89	98.01±0.15	98.25±0.54	97.93±0.35	95.34±1.23	94.89±0.93	97.63±0.29	**98.47±0.48**
Kappa	92.67±1.26	93.38±1.84	90.38±2.95	95.00±2.24	92.40±1.35	95.04±0.27	98.12±1.70	**98.25±1.87**	95.00±1.94	92.14±1.73	96.40±0.54	97.75±1.56



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 8. Visualization results of the PaviaU dataset across various models. (a) False-color image. (b) Ground-truth map. (c) MambaHSI. (d) MambaHSI+. (e) HypraMamba. (f) HSIMFormer. (g) DBMGNet. (h) Ours.

4) *Mamba-Based Methods:* MambaHSI [41] [TGRS 2024], IGroupSS [48] [TGRS 2024], S2Mamba [46] [TGRS 2025], MambaHSI+ [70] [TGRS 2025], GMambaHSI [67] [Neurocomputing 2026], and HypraMamba [69] [TGRS 2026].
5) *Hybrid Architectures Methods:* HSI-MFormer [50] [TGRS 2025] and DBMGNet [51] [TGRS 2025].

B. Comparison With State-of-the-Art Methods

We conducted a comprehensive evaluation of the proposed method from both quantitative and qualitative perspectives, comparing it with 11 state-of-the-art HSI classifiers. For the classification map visualizations, we select the most representative and latest methods among the compared approaches, including the foundational baseline MambaHSI, recent state-of-the-art Mamba variants, and representative Hybrid architectures.

1) PaviaU: Table II presents performance on the Pavia University dataset. Our method achieves top results with 98.24% OA and 98.47% average accuracy (AA), while ranking third in Kappa at 97.75%. Hybrid models GMambaHSI and MambaHSI+ achieve slightly higher Kappa, confirming that different architectures have complementary strengths. Nevertheless, our method's superiority in both OA and AA demonstrates that our heterogeneous multibranch design provides more balanced and robust classification across all categories.

Fig. 8 visualizes classification results. Existing methods exhibit misclassification between categories like bricks versus gravel, and bare soil versus meadows. In contrast, our method resolves these issues with higher intraclass consistency and precise boundaries between different land-cover categories.

This validates that our 3-D axial compression preserves spatial–spectral context to mitigate mixed-pixel uncertainty, while orientation-separated Mamba modules capture subtle differential patterns in scenarios involving both intraclass spectral variability and interclass spectral similarity.

2) LongKou: Table III presents performance on the LongKou dataset. Our method achieves optimal results with 98.04% OA, 98.14% AA, and 98.25% Kappa. Compared to the second-best IGroupSS, we improve Kappa by 1.07%, highlighting our method's robustness in classification consistency. Versus the baseline MambaHSI, gains reach 1.12% OA, 0.98% AA, and 1.00% Kappa, validating our heterogeneous multibranch design.

Fig. 9 visualizes classification results. Existing methods exhibit varying degrees of confusion when distinguishing spectrally similar categories such as broad-leaf soybean and narrow-leaf soybean. This reveals a significant bottleneck in the discriminative capacity of spatial–spectral features in existing methods when handling similar crops. In contrast, our method achieves superior class discrimination and produces considerably clearer class boundaries.

This demonstrates that the coordinated design of SMM, AP, and DE effectively captures fine-grained morphological differences between spectrally similar vegetation categories, achieving a substantial breakthrough in fine-grained agricultural subcategory classification.

3) HongHu: Table IV presents performance on the WHU-Hi-HongHu dataset. Our method achieves optimal results with 95.43% OA, 96.30% AA, and 95.90% Kappa. While the OA

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TABLE III
QUANTITATIVE RESULTS (ACC% ± STD%) OF LONGKOU

Method	GiGCN	CLONL	GSC-VIT	MambaHSI	IGroupSS	S2Mamba	MambaHSI+	GMambaHSI	HypraMamba	HSIMFormer	DBMGNet	Ours
Corn	98.06±0.71	96.38±4.65	97.20±2.85	99.49±0.36	99.21±0.40	95.81±5.94	99.39±0.45	99.33±0.36	98.79±0.94	99.34±0.56	99.48±1.00	**99.64±0.44**
Cotton	96.76±2.17	93.69±13.25	95.73±2.67	98.08±1.23	98.21±0.77	78.75±30.91	98.78±1.14	**99.29±0.45**	96.86±4.27	96.83±1.12	98.70±1.19	99.03±0.90
Sesame	97.84±0.93	70.09±31.51	91.94±14.93	98.35±1.60	99.41±0.41	98.26±2.16	99.16±0.81	99.28±0.31	99.60±0.41	98.99±0.96	86.50±17.84	**99.72±0.38**
Broad-leaf soybean	95.72±0.96	**99.57±0.28**	87.64±10.14	92.49±4.33	95.84±1.35	76.97±24.70	94.65±1.76	91.43±4.23	94.38±3.82	93.93±2.63	88.44±0.74	95.99±1.60
Narrow-leaf soybran	98.52±1.27	68.31±22.10	97.74±1.67	98.14±2.11	98.81±0.47	80.62±31.53	98.09±2.80	**99.59±0.58**	97.93±1.69	98.69±1.25	96.53±2.87	99.43±0.63
Rice	98.87±1.62	97.66±1.53	98.00±3.34	98.75±1.26	99.38±0.27	98.14±2.94	99.38±0.73	99.42±0.12	97.87±1.49	98.23±1.72	99.36±0.42	**99.69±0.27**
Water	99.22±0.55	99.47±0.46	99.52±0.22	99.62±0.33	99.02±0.68	**99.96±0.03**	99.01±1.12	97.82±0.19	98.21±1.23	96.93±1.28	99.95±0.10	99.12±0.78
Roads and houses	89.57±3.24	77.41±15.33	93.70±2.84	92.91±1.74	**96.27±0.61**	92.70±3.37	93.50±2.00	92.42±0.66	93.74±2.28	90.74±3.82	95.22±3.48	94.20±1.93
Mixed weed	81.78±3.38	73.30±18.70	93.25±3.78	96.61±1.28	94.12±4.82	96.56±1.84	**96.67±1.75**	96.48±4.25	95.94±1.37	92.26±3.90	90.38±4.03	96.48±1.38
OA(%)	97.00±0.32	93.78±1.94	94.71±3.06	96.92±1.43	97.84±0.50	90.42±9.31	97.47±0.55	96.06±3.33	96.85±1.17	96.21±1.15	95.55±0.36	**98.04±0.48**
AA(%)	95.15±0.28	86.22±3.66	94.97±1.55	97.16±0.80	97.81±0.52	90.86±6.12	97.63±0.47	97.12±1.23	97.03±0.63	96.22±1.12	94.95±1.99	**98.14±0.24**
Kappa	96.08±0.42	91.96±2.41	93.19±3.80	97.25±1.75	97.18±0.64	88.09±11.14	97.50±1.25	96.12±1.02	97.25±1.92	95.07±1.48	94.22±0.47	**98.25±1.50**

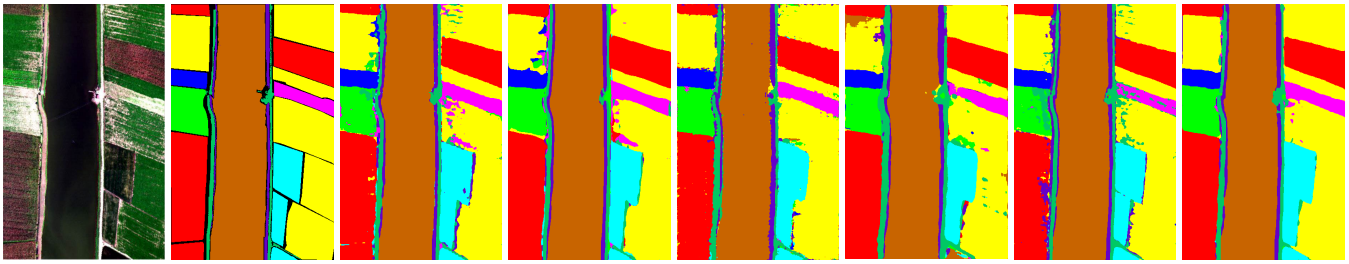


Fig. 9. Visualization results of the WHU-Hi-LongKou dataset across various models. (a) False-color image. (b) Ground-truth map. (c) MambaHSI. (d) MambaHSI+. (e) HypraMamba. (f) HSIMFormer. (g) DBMGNet. (h) Ours.

TABLE IV
QUANTITATIVE RESULTS (ACC% ± STD%) OF HONGHU

Method	GiGCN	CLONL	GSC-VIT	MambaHSI	IGroupSS	S2Mamba	MambaHSI+	GMambaHSI	HypraMamba	HSIMFormer	DBMGNet	Ours
Red roof	90.93±2.64	95.36±5.56	**98.16±1.08**	96.57±1.39	94.41±2.17	96.03±0.39	95.54±1.48	96.33±1.34	93.07±1.71	96.08±1.28	98.10±0.04	96.22±0.92
Road	51.88±7.77	78.60±6.19	73.78±9.09	95.50±2.36	92.60±2.51	96.11±0.71	95.25±2.64	96.49±1.92	86.71±5.79	94.10±2.87	**96.93±0.30**	96.71±2.19
Bare soil	78.80±2.87	**98.22±1.32**	97.21±1.81	94.62±1.35	88.05±2.89	84.95±0.31	86.57±3.84	92.59±1.73	84.84±4.19	87.21±1.61	92.19±1.34	93.29±0.99
Cotton	99.37±0.32	99.48±0.37	99.57±0.14	97.40±1.23	94.82±2.35	95.43±1.10	**99.69±0.07**	94.20±3.03	96.39±3.91	95.31±1.44	95.12±0.45	96.74±1.03
Cotton firewood	62.41±8.43	41.33±18.92	37.03±9.33	96.40±1.61	94.43±2.14	96.42±0.85	94.23±2.51	99.27±0.70	93.85±4.56	96.47±1.52	96.19±0.26	**99.28±0.80**
Rape	89.61±3.17	96.09±1.07	95.72±1.83	94.21±2.49	93.35±2.45	92.03±0.60	94.24±1.87	93.99±1.61	91.56±1.46	92.78±1.48	**96.89±0.55**	95.81±2.19
Chinese cabbage	82.91±3.07	**92.72±1.96**	90.38±3.66	86.65±3.59	82.43±4.19	77.29±0.89	83.63±5.74	84.18±2.75	73.25±5.47	80.06±4.16	78.52±0.63	89.50±2.20
Pakchoi	75.44±8.66	46.44±6.02	32.12±4.83	96.31±3.01	88.99±3.01	79.23±0.97	88.00±5.63	93.78±2.87	84.72±6.93	90.26±3.54	85.33±0.76	**97.16±2.60**
Cabbage	85.39±3.04	97.04±2.05	97.06±2.73	94.82±1.67	96.14±1.90	**99.05±0.14**	96.40±1.91	96.56±2.22	93.66±3.59	96.21±1.43	95.87±0.34	97.11±1.44
Tuber mustard	92.42±2.80	84.88±3.47	79.23±7.53	90.38±2.40	87.07±5.44	89.62±0.99	89.77±3.88	93.89±1.77	74.84±9.53	88.29±2.53	89.38±0.59	**95.15±0.92**
Brassica parachinensis	75.17±4.50	72.03±8.35	72.41±8.95	91.93±2.99	84.17±3.15	93.75±0.67	79.55±6.21	88.25±5.64	78.89±5.57	87.80±2.50	83.20±0.85	**94.85±1.85**
Brassica chinensis	82.52±5.36	66.63±8.82	65.61±8.66	87.12±5.24	84.46±5.67	89.67±1.34	91.72±3.89	90.87±4.06	67.54±7.24	80.13±8.44	88.93±0.71	**91.99±3.59**
Small Brassica chinensis	**89.28±2.70**	82.67±5.83	77.49±8.47	89.21±2.91	77.25±4.13	87.32±1.48	85.35±4.93	82.48±3.14	70.45±13.09	77.19±3.51	71.76±1.04	89.26±2.59
Lactuca sativa	68.03±5.44	87.59±7.03	86.26±11.93	93.85±1.67	90.63±4.40	96.49±0.56	90.89±4.04	**96.85±1.68**	77.14±6.88	92.14±3.45	94.42±0.65	95.81±1.82
Celtuce	21.65±2.65	75.88±12.93	63.62±20.25	98.86±1.09	98.51±1.28	99.81±0.12	87.41±7.89	99.63±0.52	93.07±6.75	98.48±1.39	99.24±0.15	**99.83±0.32**
Film covered lettuce	71.70±6.79	97.39±1.73	97.42±1.66	96.78±1.68	94.12±2.15	**99.49±0.30**	93.81±2.79	97.38±2.21	92.86±2.47	93.41±3.16	97.16±0.20	97.25±1.53
Romaine lettuce	54.94±4.04	80.09±9.70	84.92±6.09	98.69±1.39	95.32±1.75	98.05±0.69	98.24±2.58	99.05±1.64	94.39±4.24	97.25±2.18	96.24±0.23	**99.81±0.34**
Carrot	67.65±8.17	65.55±15.89	67.44±10.50	96.75±1.72	98.61±0.97	98.78±0.25	93.35±5.46	99.31±0.70	97.08±3.36	98.53±1.34	**99.50±0.22**	99.24±1.24
White radish	72.71±8.86	82.53±7.23	85.10±8.11	91.13±3.02	94.71±1.74	94.63±0.34	92.71±2.81	**95.50±1.96**	91.11±5.01	93.39±1.88	94.92±0.30	94.45±1.89
Garlic sprout	66.48±6.47	81.38±6.60	71.64±10.46	99.18±0.42	96.22±2.23	96.60±0.60	93.41±3.26	98.53±1.07	87.71±9.58	96.77±1.42	98.95±0.19	**99.26±0.47**
Broad bean	65.20±14.58	30.19±12.06	25.92±7.91	99.95±0.14	99.70±0.41	99.98±0.05	82.14±12.94	99.77±0.52	81.42±10.45	99.79±0.48	99.01±0.44	**99.99±0.02**
Tree	85.35±8.84	55.02±10.89	46.26±11.15	99.52±0.53	96.64±3.79	99.38±0.21	84.42±16.69	**99.98±0.04**	89.55±7.85	96.36±2.45	95.36±0.34	99.93±0.10
OA(%)	87.22±0.60	87.93±2.11	86.18±1.86	94.80±0.68	91.62±1.03	91.63±0.16	94.12±0.79	93.24±1.75	89.16±2.51	91.72±0.57	90.40±0.21	**95.43±0.48**
AA(%)	74.08±1.50	77.60±2.25	74.74±1.64	94.81±0.57	91.94±0.60	91.66±0.19	90.74±1.65	95.25±0.44	86.10±2.40	92.18±0.58	90.10±0.19	**96.30±0.26**
Kappa	83.95±0.73	85.06±2.49	82.93±2.16	93.90±1.91	89.53±1.23	90.24±0.19	90.14±1.73	94.81±1.10	84.14±2.22	89.65±0.68	88.81±0.25	**95.90±1.09**

improvement over the baseline MambaHSI is 0.63%, the more substantial gains in AA (1.49%) and Kappa (2.00%) highlight our method's strength in handling class imbalance—a critical challenge in this fine-grained agricultural dataset with 22 spectrally similar vegetable categories.

Fig. 10 visualizes classification results. Existing methods show confusion between cotton, cotton firewood, and trees. In contrast, our method produces homogeneous regions with clear boundaries for cotton fields, effectively distinguishing these spectrally similar vegetation types.

This demonstrates that the coordinated design of AP and DE compensates for detailed and structural information potentially overlooked by pure global sequence modeling, achieving both accuracy improvement and visual quality enhancement in this challenging agricultural scenario.

4) HanChuan: Table V presents performance on the WHU-Hi-HanChuan dataset. Our method achieves 95.42% OA, 95.44% AA, and 95.33% Kappa, showing clear superiority across all metrics. Compared to the closest-performing baseline model, we improve OA by approximately 1.94 percentage points. Against the baseline MambaHSI, gains reach 5.15%

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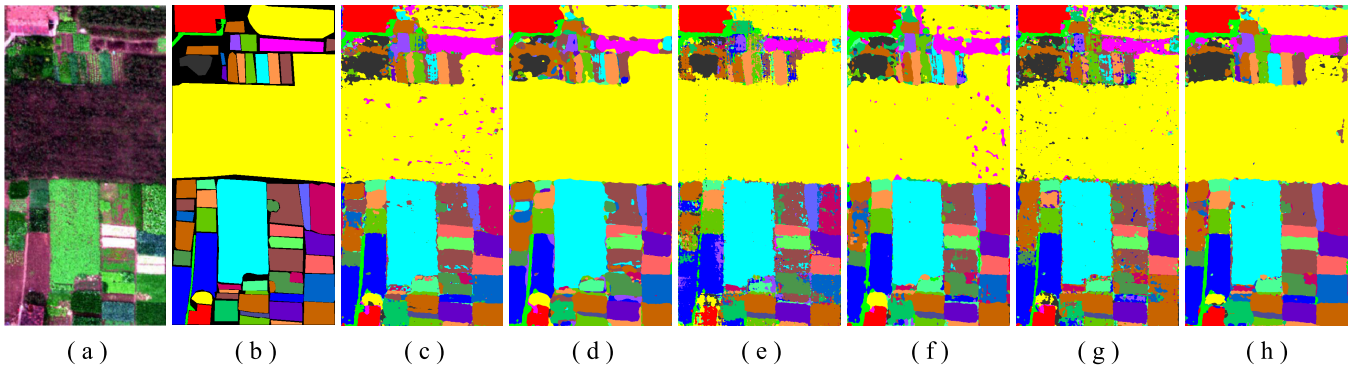


Fig. 10. Visualization results of the HongHu dataset across various models. (a) False-color image. (b) Ground-truth map. (c) MambaHSI. (d) MambaHSI+. (e) HypraMamba. (f) HSIMFormer. (g) DBMGNet. (h) Ours.

TABLE V
QUANTITATIVE RESULTS (ACC% ± STD%) OF HANCHUAN

Method	GiGCN	CLONL	GSC-VIT	MambaHSI	IGroupSS	S2Mamba	MambaHSI+	GMambaHSI	HypraMamba	HSIMFormer	DBMGNet	Ours
Strawberry	89.34±2.23	95.97±1.87	95.26±1.50	89.66±5.82	**96.74±1.01**	90.82±0.45	90.60±5.30	89.01±5.84	87.01±8.16	91.79±3.51	92.45±0.46	96.43±1.55
Cowpea	81.64±3.38	92.48±2.56	91.22±3.05	82.14±6.23	90.77±2.57	84.25±1.21	**92.63±4.70**	86.71±2.48	80.81±5.64	80.88±3.28	83.83±0.64	89.59±3.18
Soybean	81.06±6.56	80.03±9.18	78.37±7.47	94.68±3.68	88.96±3.87	93.06±0.41	91.81±4.82	90.27±5.53	90.53±11.94	91.88±3.69	93.23±0.46	**96.59±2.48**
Sorghum	67.09±4.82	91.12±7.72	85.17±12.38	98.48±1.50	97.55±1.01	96.62±0.35	98.94±1.18	**99.46±0.74**	97.76±1.34	97.43±1.59	97.57±0.17	98.96±1.05
Water spinach	34.98±2.76	56.82±12.89	46.57±13.03	99.49±0.99	98.83±0.79	99.99±0.03	99.97±0.10	99.95±0.09	95.84±4.61	99.71±0.44	99.70±0.10	**100.00±0.00**
Watermelon	53.81±9.81	38.07±8.70	35.97±7.01	75.90±5.33	88.63±4.52	92.29±0.78	84.52±6.67	76.81±4.56	76.87±6.27	84.57±3.13	87.58±0.80	**92.50±4.49**
Greens	80.84±8.87	68.87±17.46	66.90±15.67	92.57±3.79	97.12±3.76	96.29±0.31	96.28±5.38	97.82±2.38	92.18±4.99	95.20±1.00	95.83±0.48	**98.13±1.49**
Trees	89.27±4.80	81.52±7.87	82.39±5.23	71.66±6.34	84.99±3.78	73.19±0.70	90.18±4.50	70.60±9.63	70.63±11.2	76.28±4.68	72.50±0.96	**91.46±3.68**
Grass	65.14±6.79	68.59±6.61	60.64±17.62	88.97±3.78	**91.25±2.52**	84.48±1.57	83.85±5.45	90.10±3.26	78.63±12.66	84.82±5.59	82.46±0.95	90.72±3.82
Red roof	93.03±4.21	90.79±14.71	93.22±4.31	94.34±2.08	92.44±3.85	**99.47±0.07**	97.66±1.02	96.22±3.72	84.85±10.66	96.75±1.84	97.10±0.35	97.03±1.91
Gray roof	93.86±5.06	87.86±4.63	82.13±7.96	89.15±5.63	94.78±3.20	**97.48±0.51**	95.98±1.61	93.05±2.62	89.80±6.23	95.45±1.86	91.91±1.22	96.81±2.03
Plastic	89.22±7.45	63.48±14.80	51.92±6.01	89.25±6.21	88.56±4.33	97.72±0.75	87.22±14.63	95.13±3.28	86.04±6.72	92.64±4.27	94.05±0.41	**99.71±0.38**
Bare soil	66.66±6.60	50.21±6.71	53.28±10.88	79.14±3.56	84.92±4.91	79.97±0.76	86.70±2.78	77.85±4.17	77.19±5.19	77.39±5.31	80.31±1.15	**86.85±3.99**
Road	91.25±2.43	90.78±2.74	90.64±3.07	89.79±3.04	91.07±3.03	81.92±2.23	**95.92±1.32**	90.24±4.06	83.14±7.19	82.68±3.68	74.52±0.87	94.99±1.65
Bright object	46.57±17.42	62.42±10.67	70.05±12.35	94.35±2.07	96.65±2.30	99.54±0.26	98.20±1.50	96.46±2.79	88.18±2.31	92.76±2.40	**99.93±0.04**	99.32±0.71
Water	99.75±0.14	98.97±1.02	**99.77±0.23**	98.16±1.76	98.50±1.10	99.52±0.08	96.30±3.01	97.53±1.50	96.61±2.79	97.96±0.77	98.66±0.17	97.94±1.25
OA(%)	86.78±1.43	85.14±2.93	83.14±3.96	90.27±1.66	93.48±1.17	92.65±0.53	93.33±1.78	91.35±0.98	87.63±1.86	90.70±0.76	92.28±0.31	**95.42±0.73**
AA(%)	76.47±1.62	76.13±3.26	73.97±2.06	89.23±1.17	91.38±1.42	93.64±0.19	92.92±1.61	90.99±0.87	86.00±1.88	89.89±0.97	92.87±0.16	**95.44±0.59**
Kappa	82.22±1.64	82.75±3.31	80.57±4.33	88.87±1.17	91.07±1.47	90.82±0.64	92.47±3.54	90.73±2.22	84.20±2.68	89.17±0.88	90.35±0.38	**95.33±1.81**

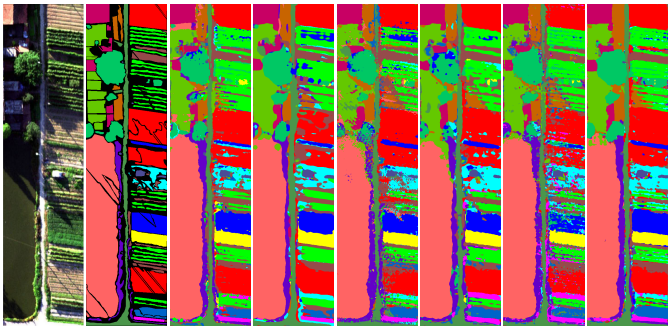


Fig. 11. Visualization results of the HanChuan dataset across various models. (a) False-color image. (b) Ground-truth map. (c) MambaHSI. (d) MambaHSI+. (e) HypraMamba. (f) HSIMFormer. (g) DBMGNet. (h) Ours.

OA, 6.21% AA, and 6.46% Kappa, validating our heterogeneous multibranch design.

Fig. 11 visualizes classification results. Existing methods struggle with key discriminations: soybean versus trees in natural vegetation, and plastic versus roads among artificial features. In contrast, our method produces homogeneous regions with sharp boundaries, accurately separating these challenging categories.

This demonstrates that our 3-D axial compression preserves spatial–spectral relationships between pixels and neighborhoods, providing richer contextual information to capture subtle spectral differences between vegetation types and distinctive spatial patterns of artificial features.

5) Houston: Table VI presents performance on the Houston dataset. Our method achieves top results with 96.20% OA, 96.88% AA, and 96.29% Kappa. Compared to the second-best MambaHSI+, we improve OA by 1.32% and Kappa by 1.58%. Versus the baseline MambaHSI, gains reach 2.54% OA, 2.28% AA, and 2.65% Kappa, validating the superiority of our multibranch heterogeneous collaboration over pure Mamba-based spatial–spectral combinations.

Fig. 12 visualizes classification results. Existing methods show severe confusion between the commercial category and features like roads, highways, and parking lot 1. In contrast, our method shows relatively better classification results in these marked regions, with clearer boundaries and reduced misclassifications.

We further analyze the commercial class, as it poses the greatest challenge among all categories. CNN-, GCN-, and Transformer-based methods on the commercial class outperform all Mamba-based methods. This suggests that distinguishing commercial from roads and parking lots relies

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TABLE VI
QUANTITATIVE RESULTS (ACC% ± STD%) OF HOUSTON

Method	GiGCN	CLONL	GSC-VIT	MambaHSI	IGroupSS	S2Mamba	MambaHSI+	GMambaHSI	HypraMamba	HSIMFormer	DBMGNet	Ours
Healthy Grass	83.21±5.03	88.24±9.00	90.16±6.48	94.82±4.52	91.40±4.62	93.28±0.05	92.70±4.49	93.62±4.19	90.23±4.40	**95.24±2.56**	94.67±1.94	93.67±4.43
Stressed Grass	82.83±7.41	93.41±5.99	95.77±2.95	96.89±3.66	91.48±5.35	97.35±2.71	94.62±2.24	94.67±2.13	93.90±4.46	96.15±3.83	97.45±1.46	**97.70±1.80**
Synthetic Grass	98.83±1.66	95.21±9.84	99.83±0.26	99.76±0.32	98.08±1.72	99.73±0.11	99.80±0.27	99.60±0.49	**99.88±0.18**	98.96±0.62	99.86±0.20	99.60±0.42
Tree	72.42±10.94	97.44±3.80	94.78±7.69	96.69±2.29	94.58±2.90	**98.85±0.24**	95.69±2.67	95.10±2.77	96.21±2.92	93.84±3.30	98.26±1.71	98.25±2.22
Soil	93.14±4.62	97.07±2.86	97.63±2.69	99.63±0.67	99.42±0.64	**100.00±0.00**	99.65±0.66	99.88±0.30	99.55±0.57	**100.00±0.00**	**100.00±0.00**	99.83±0.27
Water	88.99±6.78	94.85±4.72	96.03±2.59	98.07±1.78	97.09±2.56	92.07±0.79	98.67±2.64	98.14±2.98	98.21±1.71	98.91±1.68	**100.00±0.00**	99.26±1.15
Residential	82.34±5.24	94.02±2.94	93.11±2.39	93.21±3.18	89.28±3.44	92.00±1.57	93.89±2.88	94.12±2.75	83.52±7.56	91.11±4.16	89.27±2.16	**94.50±3.16**
Commercial	91.02±6.25	**93.80±3.94**	90.53±8.57	78.80±3.18	76.25±7.48	81.87±1.20	84.35±4.43	81.99±4.41	74.09±4.22	84.63±5.88	86.54±2.42	86.82±4.36
Road	85.51±6.43	88.32±2.83	85.20±4.96	89.83±1.92	92.48±3.02	89.86±0.56	93.51±2.26	91.00±3.30	77.73±8.77	88.54±3.47	87.05±2.58	**94.14±1.85**
Highway	94.92±2.69	76.55±6.96	85.28±4.22	96.13±1.93	95.95±3.28	94.13±0.88	**99.73±0.41**	99.26±1.22	90.34±6.47	96.03±4.00	96.88±2.86	99.65±0.56
Railway	96.52±3.78	91.57±4.40	93.08±3.33	90.52±3.74	89.51±7.38	93.40±4.64	95.44±3.11	95.66±3.64	94.45±3.25	**97.47±3.29**	94.31±3.52	96.77±2.88
Parking Lot 1	92.45±3.63	83.37±6.17	88.46±3.14	91.06±3.43	90.96±4.21	95.38±0.52	93.11±4.16	92.56±4.10	84.68±9.39	94.90±3.92	**95.64±3.26**	93.99±3.36
Parking Lot 2	88.10±5.13	89.11±4.64	90.01±4.69	97.20±2.29	89.65±5.58	98.30±1.55	95.92±3.08	96.81±2.60	93.38±3.47	96.67±2.87	95.50±0.15	**98.97±1.63**
Tennis Court	98.30±2.67	97.98±2.63	97.01±5.11	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.92±0.23	**100.00±0.00**	99.79±0.15	**100.00±0.00**
Running Track	99.14±1.89	96.08±3.96	97.73±1.23	96.37±2.12	89.29±4.02	**100.00±0.00**	96.10±2.28	96.21±2.44	99.94±0.19	99.97±0.10	**100.00±0.00**	**100.00±0.00**
OA(%)	88.13±2.26	90.31±1.73	91.85±1.05	93.66±0.99	91.67±2.30	94.44±0.47	94.88±0.87	94.42±0.98	90.11±1.86	94.63±0.93	94.83±0.87	**96.20±0.92**
AA(%)	89.85±1.91	91.80±1.46	92.97±0.92	94.60±0.92	92.36±1.94	95.08±0.42	95.55±0.80	95.16±0.91	91.74±1.44	95.50±0.77	95.68±0.69	**96.88±0.76**
Kappa	87.16±2.44	89.52±1.87	91.18±1.14	93.64±1.93	90.43±3.00	93.99±0.51	94.71±2.22	93.71±1.65	91.50±2.34	94.20±1.01	94.41±0.94	**96.29±1.80**

TABLE VII
AVERAGE PERFORMANCE COMPARISON OF DIFFERENT ARCHITECTURES ON FIVE DATASETS

base	GCN	CNN	Transformer	Mamba						Hybrid Architecture		Ours
Method	GiGCN	CLONL	GSC-VIT	MambaHSI	IGroupSS	S2Mamba	MambaHSI+	GMambaHSI	HypraMamba	HSIMFormer	DBMGNet	Ours
OA(%)	90.72±4.38	90.43±4.30	89.71±4.75	94.28±2.41	93.76±2.59	93.07±2.40	95.56±2.03	94.46±2.25	91.47±3.48	93.45±2.31	94.07±2.60	**96.67±1.30**
AA(%)	85.70±8.85	85.06±7.94	85.71±9.04	94.33±2.92	93.89±2.59	93.85±2.80	95.02±3.02	95.29±2.66	91.24±4.78	93.74±2.74	94.25±2.89	**97.05±1.14**
Kappa	90.42±5.47	88.53±4.91	87.65±5.52	93.73±3.47	92.12±3.01	91.64±3.60	94.59±3.37	94.72±3.13	90.42±5.66	92.05±3.27	92.84±3.19	**96.70±1.13**

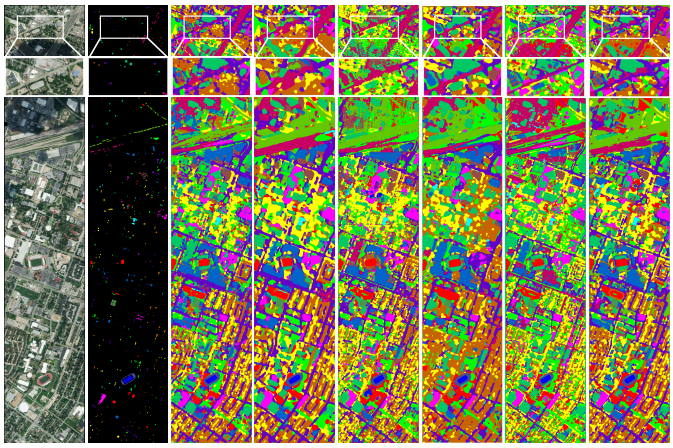


Fig. 12. Visualization results of the Houston dataset across various models. (a) False-color image. (b) Ground-truth map. (c) MambaHSI. (d) MambaHSI+. (e) HypraMamba. (f) HSIMFormer. (g) DBMGNet. (h) Ours.

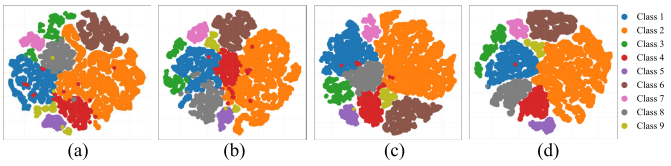


Fig. 13. Feature visualization via t-SNE. (a) SMM only. (b) SMM + AP. (c) SMM + DE. (d) Full model (SMM + AP + DE).

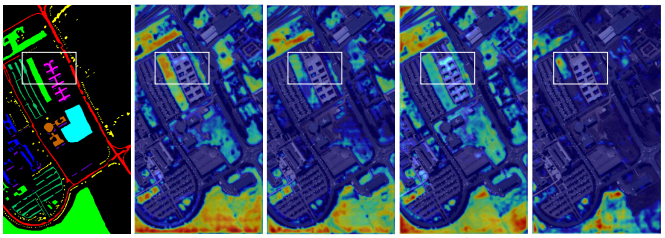


Fig. 14. Grad-CAM visualization for the meadows category in the PaviaU dataset. (a) Ground-truth. (b) Full model. (c) Addition. (d) Unidirectional Mamba. (e) Standard convolution.

more on local features than long-range dependencies. Among Mamba-based methods, ours achieves the best result of 86.82%. This indicates that our DE and AP branches effectively compensate for Mamba's weakness in local feature modeling.

This indicates that while current models disrupt inherent data structure through forced 1-D transformation and decoupled processing, our 3-D axial compression preserves global contextual relationships while providing structured feature representations, achieving a substantial breakthrough in discriminating complex urban land-cover categories.

a) Overall performance comparison: Table VII presents average performance across five datasets. Our method achieves state-of-the-art results (bold values), with improvements of 1.11% OA, 1.76% AA, and 1.98% Kappa over the second-best method (underlined). This validates the effectiveness and robustness of our structure-preserving paradigm.

C. Ablation Study

To validate the core modules' effectiveness, we conducted systematic ablation studies across five hyperspectral datasets, with results presented in Tables VIII–X and Figs. 13 and 14

1) Branch Contribution Analysis: Table VIII details branch contributions. SMM achieves competitive results

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TABLE VIII
ABLATION STUDY OF KEY COMPONENTS IN OUR METHOD

SMM	AP	DE	PaviaU	LongKou	HongHu	HanChuan	Houston
✓			92.17±1.10	96.12±1.63	92.13±1.01	93.43±1.62	90.26±0.82
✓	✓		93.62±1.24	96.50±1.46	92.35±1.00	93.88±1.18	91.36±0.80
✓		✓	97.39±1.98	98.00±1.50	95.38±0.61	94.42±0.66	96.11±0.84
✓	✓	✓	**98.26±0.84**	**98.25±1.50**	**95.43±0.48**	**95.42±0.73**	**96.19±0.94**

TABLE IX
IMPACT OF KEY COMPONENTS ACROSS FIVE HYPERSPECTRAL DATASETS. WAC, HAC, SAC, AND SM DENOTE WIDTH–AXIS COMPRESSION, HEIGHT–AXIS COMPRESSION, SPECTRAL–AXIS COMPRESSION, AND SHARE MAMBA

Setting	PaviaU	LongKou	HongHu	HanChuan	Houston
Full Model	**98.26±0.84**	**98.11±0.39**	**95.43±0.48**	**95.42±0.73**	**96.19±0.94**
w/o WAC	97.61±1.14	98.05±0.37	94.89±0.42	94.17±0.60	95.64±0.73
w/o HAC	97.17±1.98	97.81±0.51	93.67±0.72	93.48±0.68	95.91±0.81
w/o SAC	97.69±1.88	98.07±0.44	95.39±0.66	95.37±0.78	96.16±0.78
w/o SM	98.25±0.69	97.88±0.59	94.31±1.04	95.38±0.47	95.62±0.73

TABLE X
IMPACT OF KEY COMPONENTS ACROSS FIVE HYPERSPECTRAL DATASETS. PAF, BM, AND DWC DENOTE PARAMETERIZED ADAPTIVE FUSION, BIDIRECTIONAL MAMBA, AND DEPTHWISE CONVOLUTION, RESPECTIVELY

Setting	PaviaU	LongKou	HongHu	HanChuan	Houston
Full Model	**98.26±0.84**	**98.11±0.39**	**95.43±0.48**	**95.42±0.73**	**96.19±0.94**
w/o PAF	97.33±0.88	98.03±0.39	95.27±0.49	95.28±0.80	95.47±0.67
w/o BM	97.46±1.37	98.08±0.38	95.41±0.62	95.20±1.16	96.11±0.93
w/o DWC	96.72±1.37	97.71±1.00	92.23±3.32	94.13±1.50	96.04±0.94

(90.26%–96.12% OA). Adding AP improves average OA by 0.8%, showing mid-range patterns complement global modeling. Adding DE yields approximately 3.5% improvement, confirming local features compensate for detail loss. When all three branches fuse adaptively, optimal and stable performance is achieved (95.42%–98.26% OA, all std < 1%), confirming high functional complementarity.

Fig. 13 visualizes feature distributions via t-SNE. Taking Pavia University as an example, as modules are incrementally added, the fourth category (red dots) shows aggregation progress: with only SMM, samples scatter widely; adding AP and DE progressively converges outliers into the main cluster, reducing confusion. This demonstrates effective feature aggregation into compact, well-separated clusters.

2) Component Importance Analysis: Table IX presents the impact of axial compression components and parameter sharing within SMM. Removing any compression axis (WAC, HAC, and SAC) decreases OA across all datasets, validating their necessity. Height–axis compression is more critical for vertically oriented scenes (HongHu and HanChuan), while width–axis compression matters more for horizontally oriented Houston. The smaller impact of removing spectral–axis compression confirms that joint spatial–spectral modeling is essential. Removing parameter sharing (w/o SM) causes only minor accuracy drops.

Table X shows component contributions. Removing any component decreases OA across datasets, validating the necessity of the full architecture. The most significant drop occurs when removing depthwise convolution (OA declines by 3.2% on HongHu), strongly validating its critical role in distinguishing spectrally similar crops in fine-grained agricultural scenes.

Fig. 14 visualizes Grad-CAM attention for meadows in Pavia University. The complete model accurately covers meadow areas. Replacing parameterized fusion with simple summation weakens meadow attention, indicating ineffective multibranch integration. Using unidirectional Mamba erroneously shifts attention to adjacent metal roofs, revealing limited contextual perception. Replacing depthwise separable convolution with standard convolution nearly eliminates meadow attention, confirming its inability to capture discriminative spatial patterns.

Both quantitative results and visual evidence confirm that our core components collectively ensure the model consistently attends to semantically relevant regions with high accuracy and confidence. The absence of any component compromises the attention mechanism, directly degrading classification performance.

D. Complexity Analysis

Due to hyperspectral data's large-scale and high-dimensional nature, computational efficiency is crucial. We compare our method with representative approaches in model parameters, FLOPs, and inference time.

Table XI presents complexity and performance comparisons. Our method achieves optimal accuracy with exceptional efficiency. Versus Transformer-based methods (GSC–ViT), we reduce computation by orders of magnitude with thousands of times faster inference. Compared to S^2Mamba, we achieve higher accuracy with better efficiency: 0.44 versus 0.13 M parameters, 20.13 versus 48.26 GFLOPs, and 0.01 versus 10.59 s inference (three orders faster). Versus DBMGNet, we use one-third parameters and one-fifth computation with 5000× faster inference. Compared to MambaHSI, we improve accuracy while reducing computation by approximately 49% and tripling inference speed.

Table XII presents a model efficiency comparison on the Pavia University and LongKou datasets. Our method achieves the highest accuracy on both datasets while using substantially fewer computational resources. Compared to MambaHSI+, we reduce GFLOPs by 85.7% on PaviaU and 80.7% on LongKou, and cut training memory by 72.6% and 72.1%, respectively. Versus the baseline MambaHSI, we improve OA by 2.49% and 1.12% while nearly halving the GFLOPs. This validates that our heterogeneous multibranch design achieves superior accuracy–efficiency tradeoffs across different scenes.

Table XIII details the parameter and FLOPs distribution across components. The three Mamba blocks dominate parameters (92.94%), with SMM accounting for 26.14% of each block while AP and DE use only 0.80% and 4.04%, respectively. In terms of FLOPs, the embedding layer consumes 28.09% due to full-resolution processing, while DE contributes 36.79% in the first block for dense local feature extraction.

These results break the accuracy-efficiency tradeoff. By preserving data structure and optimizing resources, we

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TABLE XI

COMPARISON OF MODEL EFFICIENCY IN TERMS OF PARAMETERS, GFLOPS, AND INFERENCE TIME PER IMAGE. (MEASURED ON THE PAVIA UNIVERSITY DATASET)

Model	Params (M)	GFLOPs	Inference Time (s)	OA (%)	AA (%)	Kappa
GSC-ViT	0.15	2026.30	49.44	92.65±2.30	91.89±1.73	90.38±2.95
MambaHSI	0.41	39.18	0.03	95.77±0.89	95.86±1.12	95.00±2.24
S2Mamba	0.13	48.26	10.59	96.23±0.21	98.01±0.15	95.04±0.27
DBMNet	1.37	99.70	49.75	97.28±0.41	97.63±0.29	96.40±0.54
ours	**0.44**	**20.13**	**0.01**	**98.26±0.84**	**98.46±0.48**	**97.75±1.56**

TABLE XII

COMPARISON OF MODEL EFFICIENCY IN TERMS OF PARAMETERS, GFLOPS, AND MEMORY FOOTPRINT. (MEASURED ON THE PAVIAU AND LONGKOU DATASETS)

PaviaU	Params (M)	GFLOPs	Infer Mem (MB)	Train Mem (MB)	OA (%)	AA (%)	Kappa
MambaHSI	**0.41**	39.18	3442	6114	95.77±0.89	95.86±1.12	95.00±2.24
MambaHSI+	1.63	140.70	3652	12867	97.99±0.74	98.25±0.54	98.12±1.70
HPMamba	3.00	314.69	8060	21016	93.61±2.02	95.34±1.23	97.75±1.56
ours	0.44	**20.13**	**907**	**3523**	**98.26±0.84**	**98.46±0.48**	**97.75±1.56**
LongKou	Params (M)	GFLOPs	Infer Mem (MB)	Train Mem (MB)	OA (%)	AA (%)	Kappa
MambaHSI	**0.43**	50.96	3791.16	6622.51	96.92±1.43	97.16±0.80	97.25±1.75
MambaHSI+	1.65	158.65	4011.82	13783.66	97.47±0.55	97.63±0.47	97.50±1.25
HPMamba	3.30	343.21	8687.51	22408.41	96.85±1.17	97.03±0.63	97.25±1.92
ours	0.46	**30.66**	**1102.22**	**3850.78**	**98.04±0.48**	**98.14±0.24**	**98.25±1.50**

TABLE XIII

COMPUTATIONAL COMPLEXITY BREAKDOWN OF EACH FUNCTIONAL MODULE IN HMMAMBA ON THE PAVIA UNIVERSITY DATASET

our		Params (M)	Proportion	FLOPs	Proportion
Embedding		13.57 K	3.04%	5.65 G	28.09%
Mamba	Block0	138.5 K	30.98%	10.32 G	51.25%
	Block1	138.5 K	30.98%	2.83 G	14.04%
	Block2	138.5 K	30.98%	832.04 M	4.13%
cls_head		17.93 K	4.01%	467.88 M	2.32%
Block0	DE	18.05 K	4.04%	7.41 G	36.79%
	AP	3.59 K	0.80%	1.54 G	7.65%
	SMM	116.87 K	26.14%	1.37 G	6.82%
SMM	Share Mamba	116.48 K	26.14%	326.86 M	1.62%
	Share proj	256	0.06%	1.46 M	0.01%
	WAC	116.74 K	26.12%	381.41 M	1.89%
	HAC	116.74 K	26.12%	381.41 M	1.89%
	SAC	124	0.03%	281.23 M	1.4%

achieve dual superiority in feature discrimination and efficiency, providing a high-precision, low-complexity solution for resource-constrained applications.

V. CONCLUSION

We propose HMMamba, a lightweight heterogeneous architecture for HSI classification. Three branches—SMM, AP, and DE—capture global dependencies, medium-range context, and local details, respectively. Adaptive fusion integrates these features, achieving state-of-the-art performance with extreme efficiency (0.44 M parameters, 0.01 s inference) and strong generalization across agricultural and urban scenarios. Comprehensive interpretability analyses further validate the feature hierarchy and attention mechanisms underlying our design. Future research will focus on: enhancing 3-D axial compression through dynamic weight allocation and hierarchical strategies for ultrahigh-resolution data; extending to few-shot and semi-supervised scenarios via meta-learning and contrastive learning; and exploring multisource data fusion to advance hyperspectral classification toward more versatile multimodal remote sensing applications.

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